

Investigating a SharePoint Compromise: IR Tales from the Field

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Content

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Executive summary

Rapid7's [Incident Response](#) team recently investigated a Microsoft Exchange service account with domain administrator privileges. Our investigation uncovered an attacker who accessed a server without authorization and moved laterally across the network, compromising the entire domain. The attacker remained undetected for two weeks. Rapid7 determined the initial access vector to be the exploitation of a vulnerability, [CVE 2024-38094](#), within the on-premise SharePoint server.

Exploitation for initial access has been a common theme in 2024, often requiring security tooling and efficient response procedures to avoid major impact. The attacker's tactics, techniques, and procedures (TTPs) are showcased in this blog, along with some twists and turns we encountered when handling the investigation.

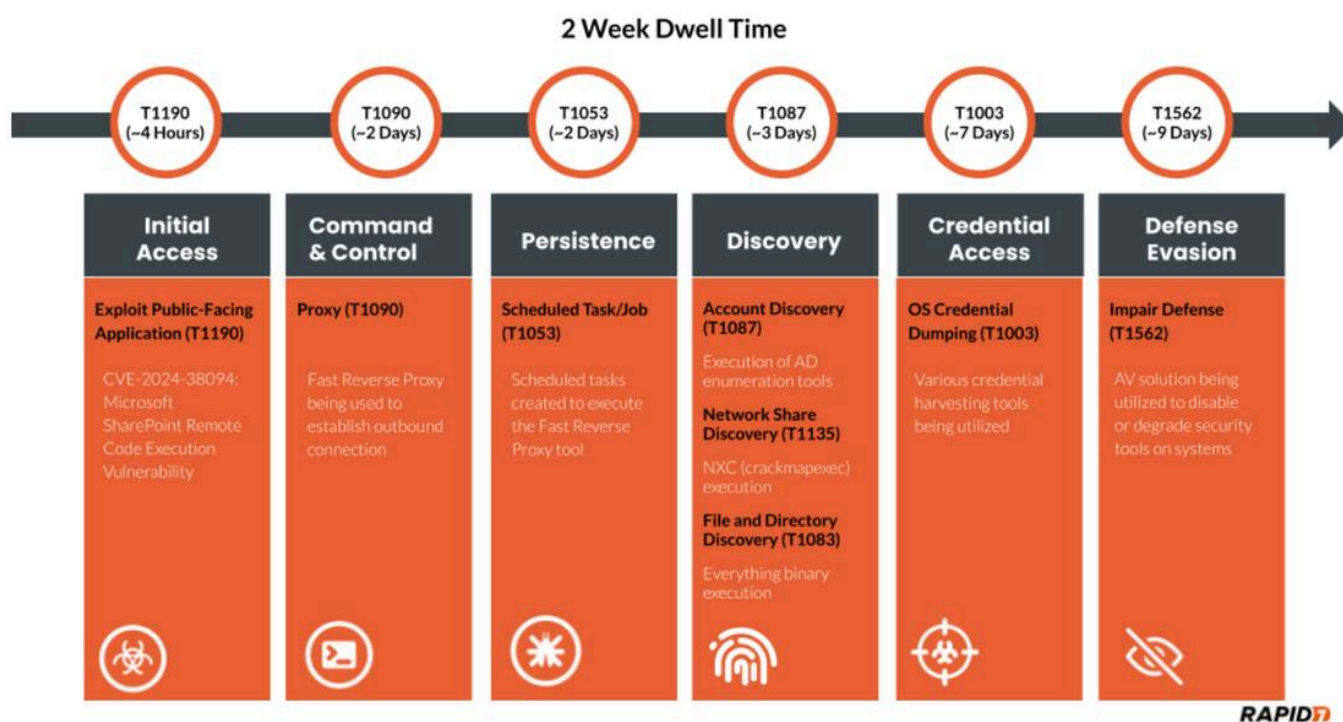
Observed attacker behavior

Rapid7 began exploring suspicious activity that involved process executions tied to a Microsoft Exchange service account. This involved the service account installing the Horoung Antivirus (AV) software, which was not an authorized software in the environment. For context, Horoung Antivirus is a popular AV software in China that can be installed from Microsoft Store. Most notably, the installation of Horoung caused a conflict with active

security products on the system. This resulted in a crash of these services. Stopping the system's current security solutions allowed the attacker freedom to pursue follow-on objectives thus relating this malicious activity to Impairing Defenses (T1562).

Zooming out from the specific event to look at the surrounding activity paints a clear picture of the attacker's intended goal. Shortly before installing Horoung AV, the attacker used Python to install Impacket from GitHub and then attempted to execute it. Impacket is a collection of open-source Python scripts to interact with network protocols, typically utilized to facilitate lateral movement and other post-exploitation objectives. The system's security tooling blocked the Impacket execution, which led to the download via browser and installation of this AV product to circumvent defenses.

As with many incident response investigations, identified clues are not always chronological, thus requiring a timeline to be constructed to understand the narrative. We must attempt to discover how the attacker compromised the system or accessed the environment in the first place. In this specific investigation, the attacker had a dwell time of two weeks. The attacker's actions are detailed chronologically in the figure below.



A great resource for identifying lateral movement involves analysis of authentication event logs from the domain controllers, specifically event ID 4624. Evidence indicated that malicious activity for this compromised Exchange service account involved more than just this single system. The source of unauthorized activity went back a week prior on a domain controller.

Analysis of the domain controller revealed that the attacker used this Exchange service account to authenticate via Remote Desktop Protocol (RDP). The attacker went on to disable Windows Defender Threat Detection (WDTD) on the system and added an exclusion for a malicious binary called msrvp.exe using the GUI. The malicious binary was placed in the C:\ProgramData\VMware\ folder but was not related to VMware. This binary is a tool called Fast Reverse Proxy (FRP), which allows external access to the system through

a NAT-configured firewall. The FRP tool requires an .ini file to provide the necessary network configuration to establish an outbound connection. The .ini file's external IP address has been provided in the Indicators of Compromise (IoCs) table in this blog post. Persistence was established for the FRP via scheduled tasks on the domain controller. Review of the C:\ProgramData\VMware\ folder used by the attacker revealed additional malicious binaries such as ADEplorer64.exe, NTDSUtil.exe, and nxc.exe. These tools were utilized to map the Active Directory environment, gather credentials, and scan systems.

Further analysis of authentication events from the domain controller indicated this malicious activity was sourced from a public-facing SharePoint server. Evidence indicated that the attacker executed Mimikatz, and there were signs of log tampering on the SharePoint server. It also indicated that a majority of system logging was disabled, and several key event log sources were absent during the investigation timeframe. Mimikatz has the ability to clear event logs and disable system logging. These malicious executions were tied to the local administrator account on the system. This would provide the necessary privileges for log tampering on the SharePoint server. However, some logs were spared, such as RDP log evidence. This indicated all authentication for the local administrator account was sourced from the local system to the local system during the in-scope time frame. The authentication information indicated that the potential initial access vector (IAV) would be tied to this SharePoint server. In light of this evidence, Rapid7 dug deeper into potential exploitation of the SharePoint services for an answer.

Rapid7 reviewed available SharePoint inetpub logs and identified the following GET and POST requests indicative of CVE-2024-38094 being exploited from the external IP address 18.195.61[.]200.

```
POST /_vti_bin/client.svc/web/GetFolderByServerRelativeUrl('/BusinessDataMetadatalCatalog/')/Files/add(url='/BusinessDataMetadatalCatalog/BDCMetadatal.bdcml)
```

```
POST /_vti_bin/DelveApi.ashx/config/ghostfile93.aspx
```

This vulnerability allows for remote code execution (RCE) on systems running Microsoft SharePoint from an external source. The proof-of-concept (PoC) code identified [here](#) was observed in available SharePoint log evidence. A great resource that explains the PoC code on Github can be found [here](#). Utilizing this vulnerability, the attacker dropped a webshell on the system. The webshell was called ghostfile93.aspx, which generated numerous HTTP POST requests from the same external IP address tied to the exploit string within log evidence. After several hours of using the webshell, the attacker authenticated into the system using the local administrator account.

Initial access occurred two weeks prior to the start of the investigation. The attacker performed other notable TTPs during the dwell time. These TTPs involved utilizing several binaries to include everything.exe, kerbrute_windows_amd64.exe, 66.exe, Certify.exe, and attempts to destroy third-party backups. The binary everything.exe can index the NTFS file system for efficient searching across files, such as recently used files and network shares. Some of the most notable binaries include 66.exe, a renamed version of Mimikatz, and

Certify.exe, which creates an ADFS certificate to utilize for elevated actions within the Active Directory environment. The remaining binary kerbrute_windows_amd64.exe has extensive capability for brute-forcing Active Directory Kerberos tickets. The attacker failed to compromise the third-party backup solution but attempted multiple methods, including access via the browser using compromised credentials and connecting over SSH.

As discussed previously, the installation of external AV products to disable security tooling was an interesting TTP identified during this investigation. Shortly after being blocked for attempted Impacket execution, Rapid7 identified the attacker leveraging an installation batch script called hrsword install.bat. The contents of this script indicate that the Huorong AntiVirus (AV) security solution was being installed. This script involved a service creation called sysdiag to execute the driver file sysdiag_win10.sys, which creates a VBS script execution parameter to execute HRSword.exe. Rapid7 observed this installation causing errors for security products on the system, potentially leading to a scenario in which the service or application would crash. These install files and all IOCs identified during this investigation have been provided in the IOC table contained within this blog.

Rapid7 customers

InsightVM and Nexpose customers can assess their exposure to the Microsoft SharePoint [CVE-2024-38094](#) with authenticated vulnerability checks added in the July 09, 2024 content release.

Rapid7 used [Velociraptor](#) during this investigation to allow for remote triage and collection of forensic artifacts on the endpoint. A Velociraptor artifact has been created to hunt for strings related to the public PoC and log evidence identified during the investigation. The artifact can be found within the Rapid7 Labs VQL Repo [here](#)

InsightIDR and Managed Detection and Response customers have existing detection coverage through Rapid7's expansive library of detection rules. Rapid7 recommends installing the Insight Agent on all applicable hosts to ensure visibility into suspicious processes and proper detection coverage. Below is a non-exhaustive list of detections that are deployed and will alert on behavior related to exploitation of this vulnerability.

Suspicious Commands Launched by Webserver IIS Launching Discovery Commands IIS Spawns PowerShell Attacker Tool - Impacket Attacker Tool - MimiKatz Attacker Technique - Hash Dumping With NTDSUtil Attacker Technique - Clearing Event Logs Defense Evasion - Disabling Multiple Security or Backup Products

Rapid7 also recommends ensuring that SharePoint is patched to the latest version.

MITRE ATT&CK techniques

| Tactic | Technique | Details | | | Initial Access | Exploit Public-Facing Application (T1190) | CVE-2024-38094: Microsoft SharePoint Remote Code Execution Vulnerability | | | Defense Evasion | Impair Defense (T1562) | AV solution being utilized to disable or degrade security tools on systems. | | | Discovery | Account Discovery (T1087) | Usage of AD enumeration tools | | | Command and Control | Proxy (T1090) | Fast Reverse Proxy being used to

establish outbound connection | | | Discovery | File and Directory Discovery (T1083) | Everything.exe being observed on in-scope systems. | | | Discovery | Network Share Discovery (T1135) | nxc.exe being observed on in-scope systems. | | | Credential Access | OS Credential Dumping (T1003) | Various credential harvesting tools observed on in-scope systems | | | Persistence | Scheduled Task/Job (T1053) | Scheduled tasks observed on in-scope systems to execute the FRP tool. | |

Indicators of Compromise

Attribute	Value	Description
Filename and Path	c:\users\Redacted\documents\everything-1.4.1.1024.x86\everything.exe	Binary to locate files
SHA256	d3a6ed07bd3b52c62411132d060560f9c0c88ce183851f16b632a99b4d4e7581	Hash for everything.exe
Filename and Path	c:\programdata\vmware\66.exe	Renamed mimikatz.exe
SHA256	61c0810a23580cf492a6ba4f7654566108331e7a4134c968c2d6a05261b2d8a1	Hash for mimikatz.exe
Filename and Path	c:\programdata\vmware\certify.exe	Creates an ADFS certificate to utilize for elevated actions within the Active Directory environment.
SHA256	95cc0b082fcfc366a7de8030a6325c099d8012533a3234edbfd555df082413c7	Hash for certify.exe
Filename and Path	c:\programdata\vmware\kerbrute_windows_amd64.exe	Used to perform Kerberos pre-auth brute forcing.
SHA256	d18aa84b7bf0efde9c6b5db2a38ab1ec9484c59c5284c0bd080f5197bf9388b0	Hash for kerbrute_windows_amd64.exe
Filename and Path	c:\programdata\vmware\msvrp.exe	Fast Reverse Proxy tool for allowing external access to the system through a NAT configured firewall.
SHA256	f618b09c0908119399d14f80fc868b002b987006f7c76adbceclac11b9208940	Hash for msvrp.exe
Filename and Path	c:\programdata\vmware\nxc.exe	Newer version of the CrackMapExec Network Pentesting tool.
SHA256	95cc0b082fcfc366a7de8030a6325c099d8012533a3234edbfd555df082413c7	Hash for nxc.exe
Filename and Path	c:\programdata\vmware\adexplorer64.exe	Active Directory Enumeration Tool
SHA256	e451287843b3927c6046eaabd3e22b929bc1f445eec23a73b1398b115d02e4fb	Hash for adexplorer64.exe
Filename and Path	c:\users\Redacted\documents\h\hrsword install.bat	Component of Huorong AV
SHA256	1beec8cecd28fdf9f7e0fc5fb9226b360934086ded84f69e3d542d1362e3fdf3	Hash for hrsword install.bat
Filename and Path	c:\users\Redacted\documents\h\hrsword.exe	Component of Huorong AV
SHA256	6ce228240458563d73c1c3cbbd04ef15cb7c5badacc78ce331848f5431b406cc	Hash for hrsword.exe
Filename and Path	c:\Windows\System32\drivers\sysdiag_win10.sys	System driver component of Huorong AV
SHA256	acb5de5a69c06b7501f86c0522d10fefa9c34776c7535e937e946c6abfc9bbc6	Hash for sysdiag_win10.sys
Log-Based IOC	POST /_vti_bin/client.svc/web/GetFolderByServerRelativeUrl('/BusinessDataMetadataC	

atalog/')/Files/add(url='/BusinessDataMetadataCatalog/BDCMetadata.bdcml | POC code identified in SharePoint logs. | | | Log-Based IOC | POST
/_vti_bin/DelveApi.ashx/config/ghostfile93.aspx | Webshell identified within SharePoint logs.
| | | IP Address | 54.255.89[.]118 | IP address from .ini file for Fast Reverse Proxy tool | | | IP Address | 18.195.61[.]200 | Source IP address from exploitation and webshell communications | |

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Noah Hemker

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